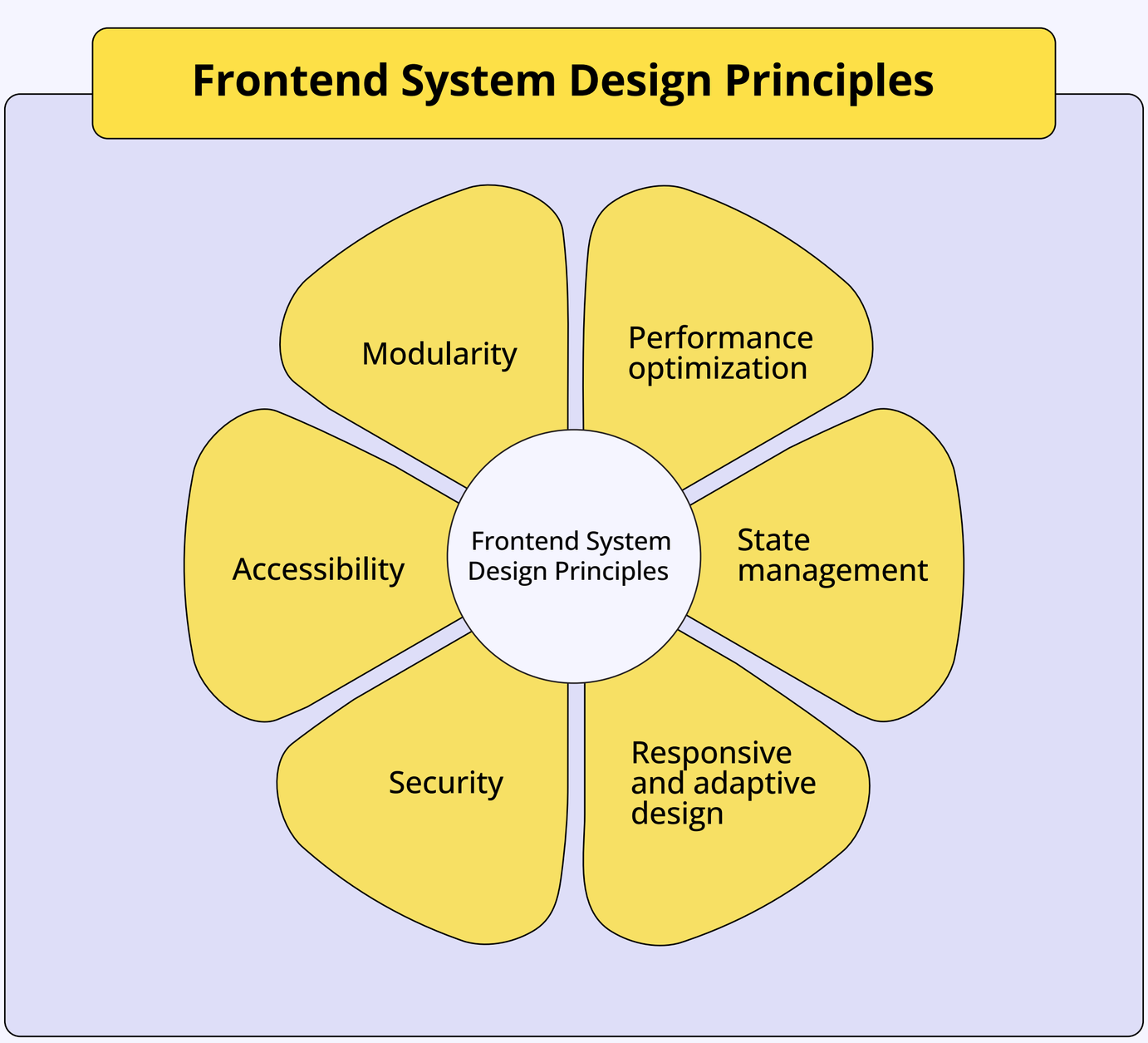
Frontend System Design Principles

What is Frontend System Design?

It is the process of architecting and planning the structure, components, and user experience of web or mobile interfaces to ensure scalability, performance, and maintainability.
These principles are the foundational guidelines that directly inform and shape the decisions made during the frontend System Design process. These principles are:

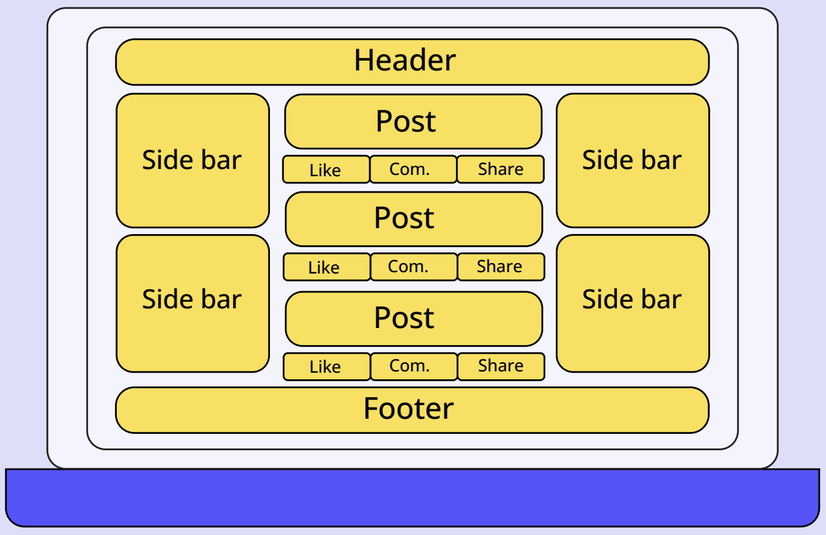


1. Modularity

It means breaking the UI into independent, reusable components to improve scalability, maintainability, and testing in frontend systems.
We can achieve modularity by:

- Using component-based frameworks (React, Vue, and Svelte).
- Following Atomic Design or Smart-Dumb (Container-Presenter) patterns.
- Isolating UI components using tools like Storybook.
- Using design systems for consistency.

Real-world example
In a social media app, Like, Comment, and Share are modular components reused across posts, reels, and feeds, each with different props like post ID or user access.



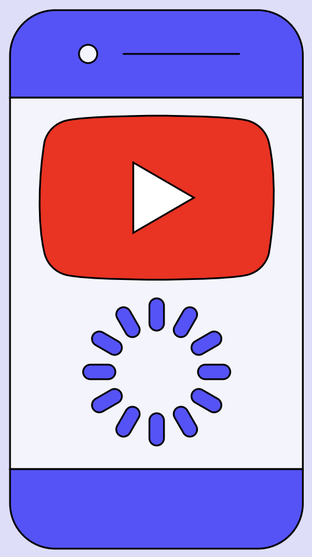
Each component is independent and reusable

2. Performance Optimization

It is about reducing load time and ensuring smooth interactions for a seamless user experience.
We can achieve performance optimization by:

- Code splitting (e.g., React.lazy).
- Lazy loading assets and images.
- Memoizing expensive operations (e.g., useMemo, memo).
- Using CDN for static assets.
- Reducing bundle size via tree shaking.

Real-world example
YouTube defers loading comments and sidebar videos until the user scrolls down or interacts.

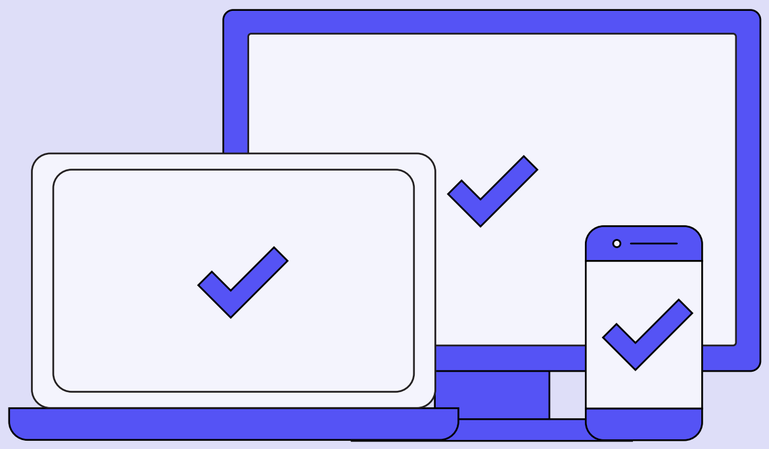


3. State management

It means handling and maintaining the state (data or information) within an application. In a frontend context, it refers to managing dynamic values, like user input, application data, and UI settings, that affect the user interface and experience.
We can achieve state management by:

- Using local state with hooks (`useState, ref`).
- Centralized state libraries (Redux, MobX, Zustand, and Pinia).
- Handling derived/computed state properly.
- Managing async state with tools like React Query or SWR

Real-world example
In Slack, typing a message updates the local state instantly while syncing with the server in the background.

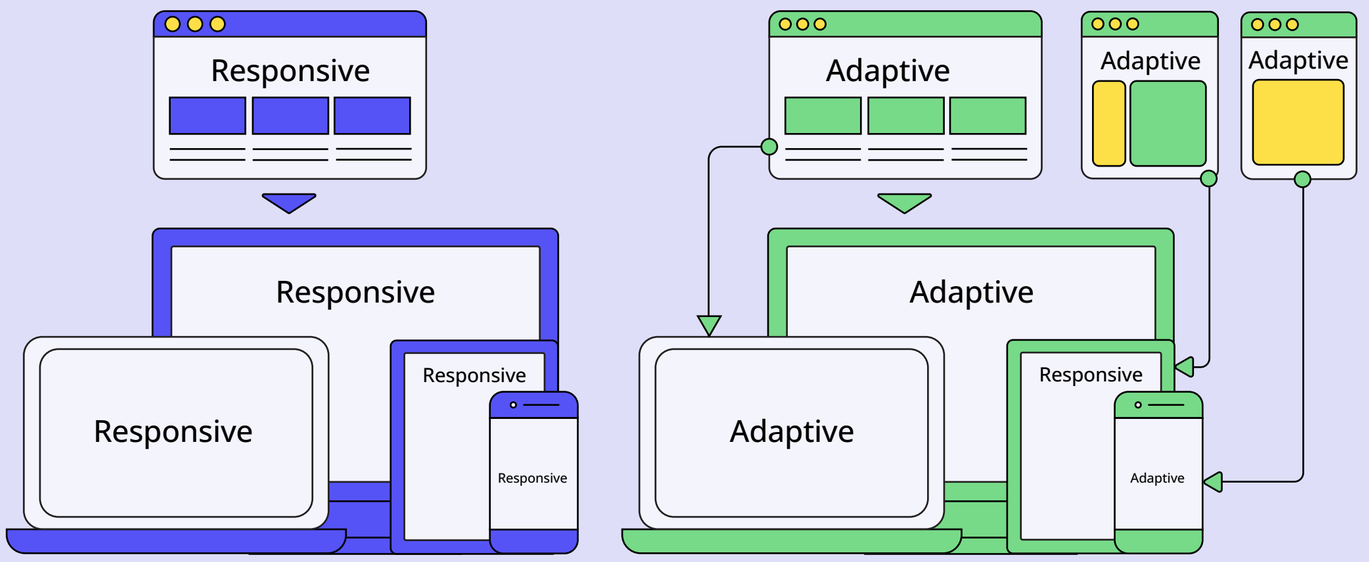


4. Responsive and adaptive design

It fluidly adjusts the UI across screen sizes using flexible layouts, grids, and media queries, while adaptive design switches between predefined layouts tailored for specific devices or screen sizes.
We can achieve responsive and adaptive design by:

- Using CSS Flexbox/Grid.
- Applying media queries and breakpoints.
- Designing mobile-first layouts.

Real-world example
Spotify adapts its layout from multi-pane desktop view to a simplified, stacked mobile interface.



5. Security

It prevents attacks and ensures safe handling of user data and sessions.
We can achieve security by:

- Escaping and sanitizing all user inputs.
- Using HTTPS, Content Security Policies (CSP).
- Preventing XSS (Cross-Site Scripting) and CSRF (Cross-Site Request Forgery) via tokens and headers.
- Storing tokens in memory or secure cookies.
- Avoiding sensitive logic exposure in the frontend.

Real-world example
Google Drive uses strong CSP headers, secure cookies, and prevents third-party scripts from injecting content.



6. Accessibility

It ensures that all users, including those who are differently-abled, can perceive, navigate, and interact with a website or application by following standards like Web Content Accessibility Guidelines (WCAG).
We can achieve accessibility by:

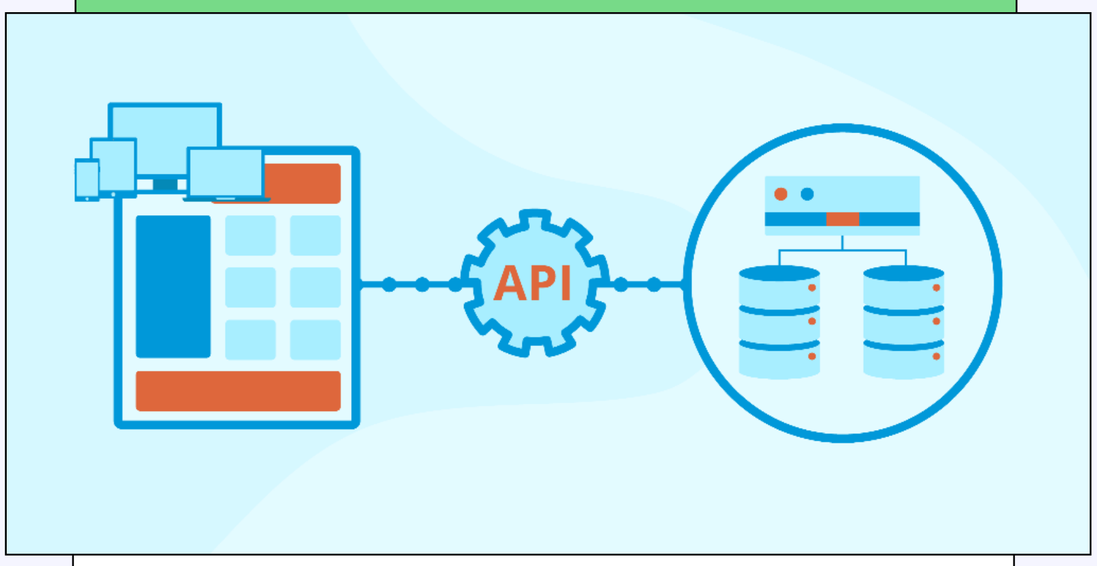
- Using semantic HTML elements (<button>, <label>, <nav>).
- Adding Accessible Rich Internet Applications (ARIA) roles and labels where needed.
- Supporting keyboard navigation.
- Ensuring sufficient color contrast.
- Testing with screen readers.

Real-world example
LinkedIn supports keyboard-only navigation, screen reader labels, and ARIA live regions for announcements.

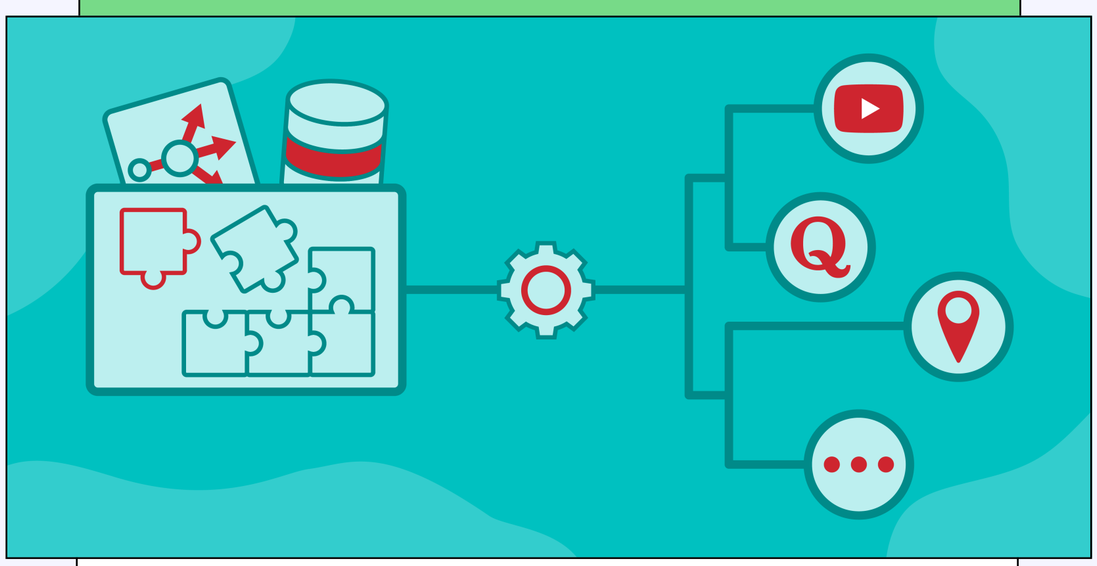


System Design Courses

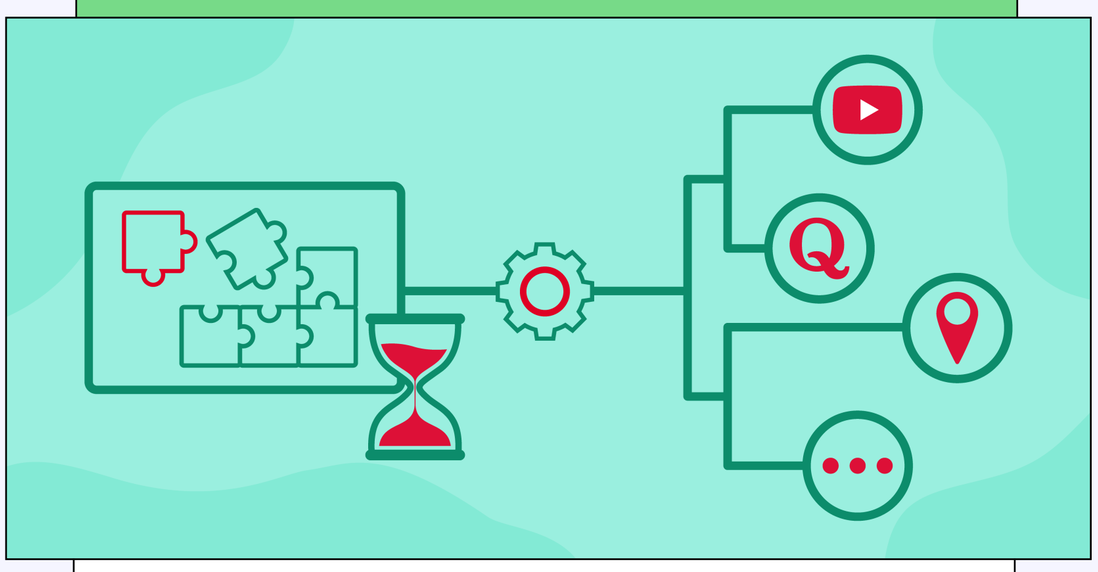
Grokking the Frontend System Design Interview



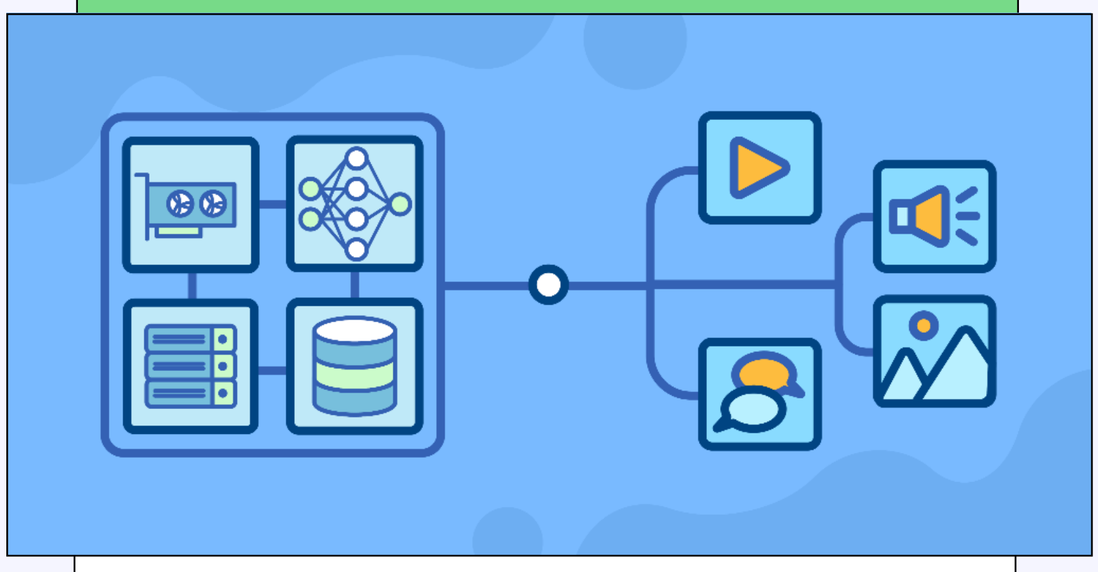
Grokking the Modern System Design Interview



System Design Interview Prep Crash Course



Grokking the Generative AI System Design



Grokking the Product Architecture Design Interview

